

Activity-dependent lateral inhibition enables ensemble synchronization of odor-activated neurons in the olfactory bulb

Tal Dalal, Rafi Haddad 

The Gonda Multidisciplinary Brain Research Center, Bar-Ilan University, Ramat-Gan, 5290002, Israel

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Abstract

Information in the brain is represented by the activity of neuronal ensembles. These ensembles are adaptive and dynamic, formed and truncated based on the animal's experience. One mechanism by which spatially distributed neurons form an ensemble is synchronizing their spiking activity in response to a sensory event. In the olfactory bulb, odor stimulation evokes rhythmic gamma activity in spatially distributed mitral and tufted cells (MTCs). This rhythmic activity is thought to enhance the relay of odor information to the downstream olfactory targets. However, how only the odor-activated MTCs are synchronized is unknown. Here, we demonstrate that light activating one set of MTCs can gamma-entrain the spiking activity of another set. This lateral synchronization was particularly effective when both MTCs fired at the gamma rhythm, facilitating the synchronization of only the odor-activated MTCs. Furthermore, we show that lateral synchronization did not depend on the distance between the MTCs and is mediated by Granule cells. In contrast, lateral inhibition between MTCs that reduced their firing rates was spatially restricted to adjacent MTCs and was not mediated by Granule cells. Our findings reveal a simple yet robust mechanism by which spatially distributed neurons entrain each other's spiking activity to form an ensemble.

Highlights

1. MTC activation entrains the spike timing of other MTCs in an activity-dependent and spatially-independent manner.
2. MTC to MTC suppression is activity- and distance-dependent
3. Spatially distributed Granule cells control MTC's spike timing, yet do not substantially affect their firing rate.

eLife assessment

This **valuable** study provides in vivo evidence for the synchronization of projection neurons in the olfactory bulb at gamma frequency in an activity-dependent manner. This study uses optogenetics in combination with single-cell recordings to selectively activate sensory input channels within the olfactory bulb. The data are thoughtfully analyzed and presented; the evidence is **solid**, although some of the conclusions are only partially supported.

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Introduction

Information in the brain is represented by the activity of ensembles of neurons, typically interconnected through a network of interneurons (Buzsáki and Chrobak, 1995). These interactions reshape the ensemble activity as it evolves in time and space. One mechanism by which spatially distributed neurons form an ensemble is by synchronizing their spiking activity in response to a sensory event (Buzsáki, 2010). This synchronization is thought to enhance the transmission of information to downstream targets (Dalal and Haddad, 2022; Macleod et al., 1998; Pritchett et al., 2015; Sohal, 2016).

The olfactory bulb (OB) is the first processing station of the olfactory system. In the OB, odors are represented by sets of activated glomeruli that reflect the responses of olfactory sensory neurons. Each glomerulus innervates a small number of MTCs. These MTCs interact with each other via a dense network of inhibitory neurons stratified throughout all six OB layers. Several studies have shown that odorants evoke strong spike gamma-entrainment in MTCs (Beshel et al., 2007; Dalal and Haddad, 2022; Fukunaga et al., 2014; Lepousez and Lledo, 2013), as well as synchronous firing of MTCs (Kashiwadani et al., 1999). This synchronization is mediated by granule cells (Schoppa, 2006)(GCs) and has recently been shown to enhance odor-information transmission to the Piriform cortex (Dalal and Haddad, 2022). However, the method by which only the odor-activated MTCs are synchronized is unknown. One prominent hypothesis is that GCs enable synchronization solely between odor-activated MTCs (Egger and Kuner, 2021; Galán et al., 2006).

GCs are the most abundant type of interneuron in the OB. They form dendrodendritic synapses with MTCs (Shepherd, 2004) and, via these synapses, can provide recurrent and long-range lateral inhibition. Earlier studies suggested that GCs mediate lateral inhibition between MTCs shaping their firing rate (Arevian et al., 2008; Giridhar et al., 2011; Yokoi et al., 1995). In contrast, more recent studies demonstrated that in-vivo lateral inhibition is sparse and weak (Fantana et al., 2008; Lehmann et al., 2016; Pressler and Strowbridge, 2017), therefore unlikely to substantially suppress MTC firing rates. One study showed that optogenetically silencing the GCs did not affect odor-evoked inhibitory responses (Fukunaga et al., 2014). These findings cast doubts on the role of GCs in suppressing MTCs' odor-evoked firing rate (Burton, 2017).

Here, we used odor and optogenetic stimulations of MTCs and GCs in anesthetized mice, to study how active MTC ensembles interact to regulate their spikes' timing and firing rates. We found that MTCs form two types of interactions that are mediated by two types of interneurons. One

interaction enables the synchronization of only odor-activated MTCs dispersed on the OB surface, and is mediated by GCs. The other type of interaction affecting MTC firing rate is limited to relatively nearby MTCs, and, contrary to the view in the field, is *not* mediated by GCs.

Results

Activity-Dependent Lateral Synchronization of MTCs

To investigate how MTCs interact we expressed the light-gated channel rhodopsin (ChR2) exclusively in MTCs by crossing the Tbet-Cre and Ai39 mice lines (Grobman et al., 2018 [DOI](#); Haddad et al., 2013 [DOI](#)), and recorded the spiking activity of MTCs in anesthetized mice during optogenetic stimulation. We first mapped each recorded cell's receptive field, i.e., the glomeruli on the dorsal OB that affect its firing rates when they are light-stimulated. Pseudo-random light patterns composed of multiple light spots were projected over the OB surface (**Supplementary Figure 1a** [DOI](#)) and the spike-triggered-average was computed to obtain the receptive field map (**Figure 1a** [DOI](#); 'STA', see Methods). Each light pattern comprised 5-10 multiple light spots of size 88-110 μm^2 . This stimulation protocol ensured the activation of varied MTC combinations associated with different glomeruli, among them the MTC we recorded from. As expected, light-stimulating the area above the recording electrode activated the recorded MTC (the 'hotspot'). This protocol also revealed regions on the OB surface that had a suppressive effect on the activity of the recorded MTC (**Figure 1a** [DOI](#)). Based on the receptive field map, we selected several spots and light-activated them either alone or simultaneously with the spot that activated the recorded MTC using four increasing light intensities (**Figure 1a-b** [DOI](#); N = 5 mice; 27 neurons; overall 127 pairs were tested, 2-8 different pairs per cell). We found that in a subset of neurons, paired light activation precisely aligned the spike times of the recorded MTC across and within trials without affecting its firing rate (**Figure 1c** [DOI](#) and **Supplementary Figure 1b** [DOI](#)), giving rise to a gamma rhythm (**Figure 1d-e** [DOI](#) and **Supplementary Figure 1c** [DOI](#)). To quantify the change in spike precision and gamma entrainment at the population level, we computed the difference in the area under the power spectrum density (PSD) curve between the two conditions at the gamma range (termed as 'delta entrainment'; 40-70Hz, see Methods). We found that the change in entrainment was significantly higher than zero (**Figure 1f** [DOI](#); $P < 0.001$ and $P = 0.13$ for real and shuffled data, respectively; two-tailed paired *t*-test, N = 319/511 values from all pairs and light intensities that significantly responded to light stimulation). Overall, we found that in ~16% of the stimulated pairs, paired activation significantly enhanced the gamma entrainment (N = 50/319 delta values that exceeded the 95% confidence interval of the shuffled distribution, colored brown in **Figure 1f** [DOI](#)). Varying the light intensities revealed that the enhanced gamma entrainment was most effective when the postsynaptic MTCs firing rate was ~40Hz (**Figure 1g** [DOI](#)). Analyzing the locations in which paired activation significantly increased the temporal precision revealed no significant correlation between the pair's distance and its gamma entrainment change, suggesting that lateral entrainment is driven by MTCs distributed on the OB surface without any discernable spatial organization (**Figure 1h** [DOI](#); $r = 0.05$, $P = 0.73$, Spearman correlation). Overall, we found activity-dependent enhancement of MTC spike precision in the gamma range that is distance-independent.

Activity-Dependent Lateral Suppression of MTCs is Confined in Space

In addition to lateral synchronization between co-active MTCs, we found that paired activation could suppress the recorded MTC firing rate (**Figure 2a-b** [DOI](#)). **Figure 2c** [DOI](#) shows the response of a recorded MTC when light-stimulated alone, and simultaneously with other MTCs under four different light intensities (pair #1 or pair #2, as marked in **Figure 2a** [DOI](#)). Activation of pair #1 evoked lateral suppression that was dependent on the recorded MTC firing rate, while activation of pair #2 did not affect the MTC firing rate across any light intensities. Plotting the evoked change

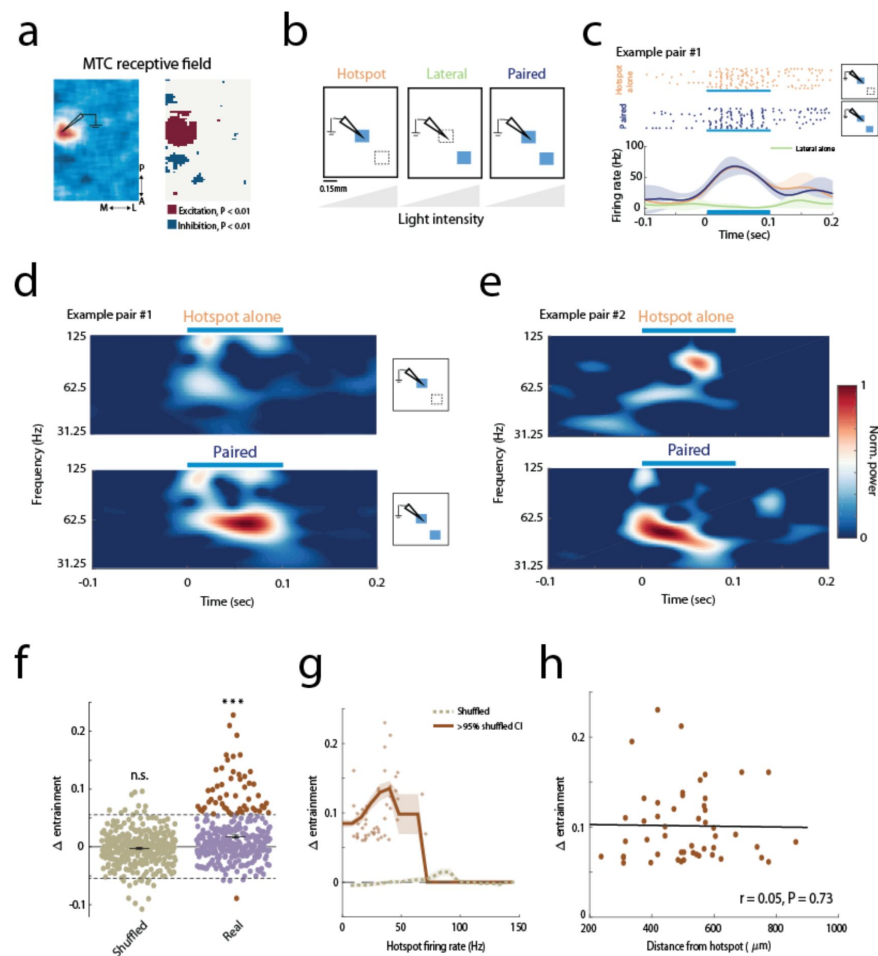


Figure 1

Activity-dependent lateral-enhancement of spike times

a) Left: STA map with an excitatory region near the electrode location and presumably several surrounding inhibitory spots. Right: the significance map ($P < 0.01$ relative to shuffled data, see Methods).

b) Schematic illustration of the experimental setups. Photo-stimulation of each spot alone (hotspot or lateral spot conditions marked by orange and green text, respectively) or paired stimulation (marked in blue) using four different light intensities.

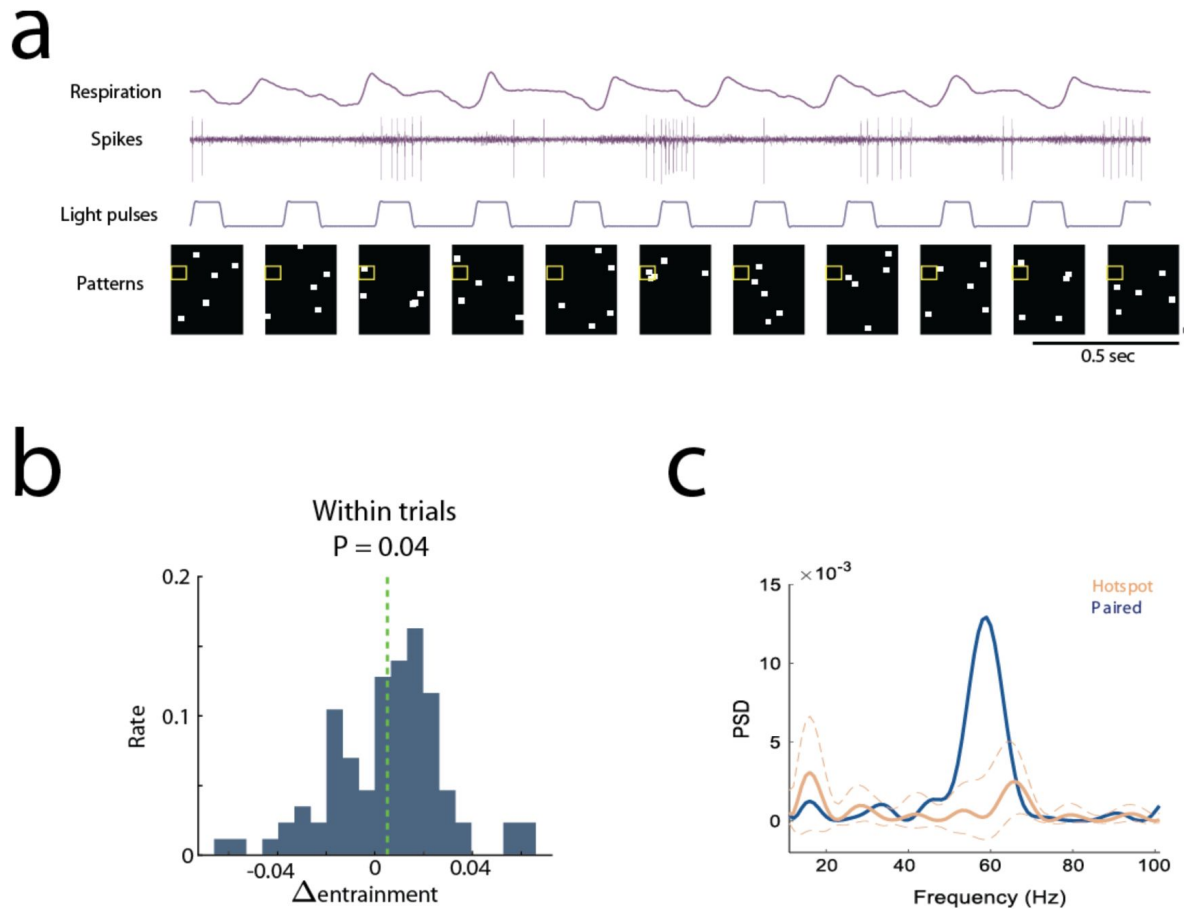
c) Raster plots and smoothed PSTHs of the response to light stimulation of the hotspot (top, orange) and paired stimulation (middle, blue). Note, the increase in spike time accuracy within and across trials when both spots are activated (middle panel). Light stimulation did not affect the average firing rate (lower panel). The effect of light stimulation of the lateral spot alone is shown in green.

d-e) Examples of MTCs time-frequency wavelet analysis from two different mice. Example pair #1 is the pair displayed in c. Both examples show a strong gamma rhythm following paired stimulation. In pair #1, gamma power peaked at ~58Hz, and pair #2 at ~48Hz.

f) Paired stimulation increased the spikes' temporal precision. Mean $\pm$ SEM of the change in spikes entrainment at the population level ($N = 319/511$ values from all pairs and light intensities that significantly responded to light stimulation, $P = 0.13$ and $P < 0.001$ for shuffled (green) and real (purple) data, respectively; two-tailed paired t -test). In brown are values that exceeded the 95% confidence interval of the shuffled data distribution values of increased and decreased spike entrainment, respectively; confidence interval is marked by dashed black lines).

g) Lateral entrainment is activity-dependent. The moving average of the data shown in f is plotted as a function of the firing rate of the postsynaptic MTC ($N = 50$ values of increased entrainment). The increase in entrainment was largest when the neuron fired at ~40Hz. The color code is the same as in f. The shuffled data is shown in a dashed green line.

h) Spike entrainment does not depend on the distance between the MTC pair. No significant correlation was found between the increase in spike-entrainment and the distance from the hotspot ($r = 0.05$, $P = 0.73$, Spearman correlation; $N = 50$ values with significant increase in spike entrainment, brown dots in g).



Supplementary Figure 1

Spike entrainment between and within trials

a) Light stimulation protocol. An image containing N randomly distributed light patches ($N = 5$ in this example) is projected on the dorsal bulb in each trial. The light patterns are shown in the bottom panel. The yellow rectangle marks the region around the recording electrode. The spiking activity and the respiratory signal are shown above. Multiplying each pattern by the firing rate it evoked and averaging across all trials gives the STA activity map (see Methods). Scale bars: 0.5 second; 110 μm .

b) Paired activation enhances the spike precision at the gamma range within each trial ($P = 0.04$, two-tailed paired t -test, see Methods). Delta entrainment was computed per trial (see Methods). The mean is marked with a dashed green line.

c) Power spectral density (PSD) during the light stimulation duration (100 ms) of the example shown in **Figure 1c-d**.

Dashed lines represent the 95% confidence interval constructed using bootstrapping.

in firing rate caused by paired stimulation as a function of the MTC firing rate when stimulated alone revealed that lateral inhibition is most effective when the recorded MTC fires in the gamma range (i.e., ~30-80 Hz, **Figure 2d**). Light stimulating the lateral spots alone did not affect the MTC baseline firing rate (**Supplementary Figure 2a**). Overall, in 19% (24/127) of the tested MTC pairs, the recorded MTC was significantly inhibited following paired stimulation. In contrast to the lack of spatial organization between MTC pairs that caused spike entrainment, we found a significant correlation between the distance of the paired spots and the level of evoked inhibition (**Figure 2e**; $r = 0.45$, $P = 0.001$, Spearman correlation). To further verify that lateral inhibition is limited to proximal MTCs, we analyzed the inhibition found in the receptive field (STA) maps. Centering all z-scored maps relative to the 'hotspot' location ($N = 27$ neurons from 5 mice) revealed that MTC-to-MTC suppressive interactions are strongest and densest in regions that are adjacent to the recorded MTC (~200-400 μm) and diminished quickly beyond that (**Figure 2f** and **Supplementary Figure 2b**). Furthermore, computing the STA maps while excluding the light patterns that contained a light spot that hit the electrode location ('hotspot' area) resulted in maps that did not contain suppressive regions (**Supplementary Figure 2c-d**). This analysis further confirms that lateral suppression requires the recorded MTC to be active. In summary, we show that MTC-to-MTC suppressive interactions are spatially-confined and activity-dependent.

Two different neural circuits mediate spike suppression and entrainment

In our experimental design, the same MTC participated in more than one pairing. Analyzing these pairs revealed that while light-stimulation of one pair could enhance the spike-precision without affecting its firing rate, activation of a different pair at the same light intensity could suppress the MTC firing rate without affecting its spike-precision (**Figure 3a**). Furthermore, light-activating the same pair with two different intensities could suppress the MTC firing rate at one intensity and enhance spike entrainment in the second (**Figure 3b**). These findings strongly suggest that spike suppression and entrainment are not features of the recorded cell type (i.e., mitral versus tufted cells) but instead reflect two different circuits which the MTCs form with each other.

GC activation increases MTC synchrony in an activity-dependent and location-independent manner

We next sought to understand the neural circuits that mediate spike entrainment and suppression. We examined whether GC-MTC interactions underlie the temporal changes or the suppressive interactions we found. We conditionally expressed ChR2 in GCs by injecting AAV5-EF1a-DIO-ChR2 to Gad2-Cre mice into the GC layer (**Figure 4a**) as in (Dalal and Haddad, 2022; Fukunaga et al., 2014). We used odor stimuli to activate the recorded MTC. To activate all GCs belonging to a glomerulus column we used a relatively large light spot sized ~330 μm^2 (Egger and Urban, 2006). We then tested how light-activation of subsets of GCs at different locations affects MTC odor-evoked temporal dynamics and spike rate suppression (**Figure 4b**; $N = 4$ mice, $N = 31$ cell-odor pairs from 22 cells). To examine the temporal effects, for each cell-odor pair we extracted the MTC odor-evoked spike phases from within the LFP-gamma cycles across all trials. We then quantified the level of spike LFP-gamma coupling using the pairwise phase consistency measure (PPC1). This measure serves to minimize the bias of the neuron's firing rate on the synchrony level (Vinck et al., 2012). Only MTCs that responded to odor by excitation were analyzed ($N = 18/31$, $P < 0.05$, two-tailed paired t -test). We found that light-activating GC columns significantly increased MTC spike phase-locking (**Figure 4c**, $N = 3$ light spots for each cell-odor pair, a total of 54 light spots tested, $P = 0.0016$, two-tailed paired t -test). Furthermore, the increased phase-locking depended on the level of the MTC firing rate and did not significantly correlate with the distance from the MTC (**Figure 4d-e**). Plotting the odor-evoked spikes by aligning them to an arbitrary

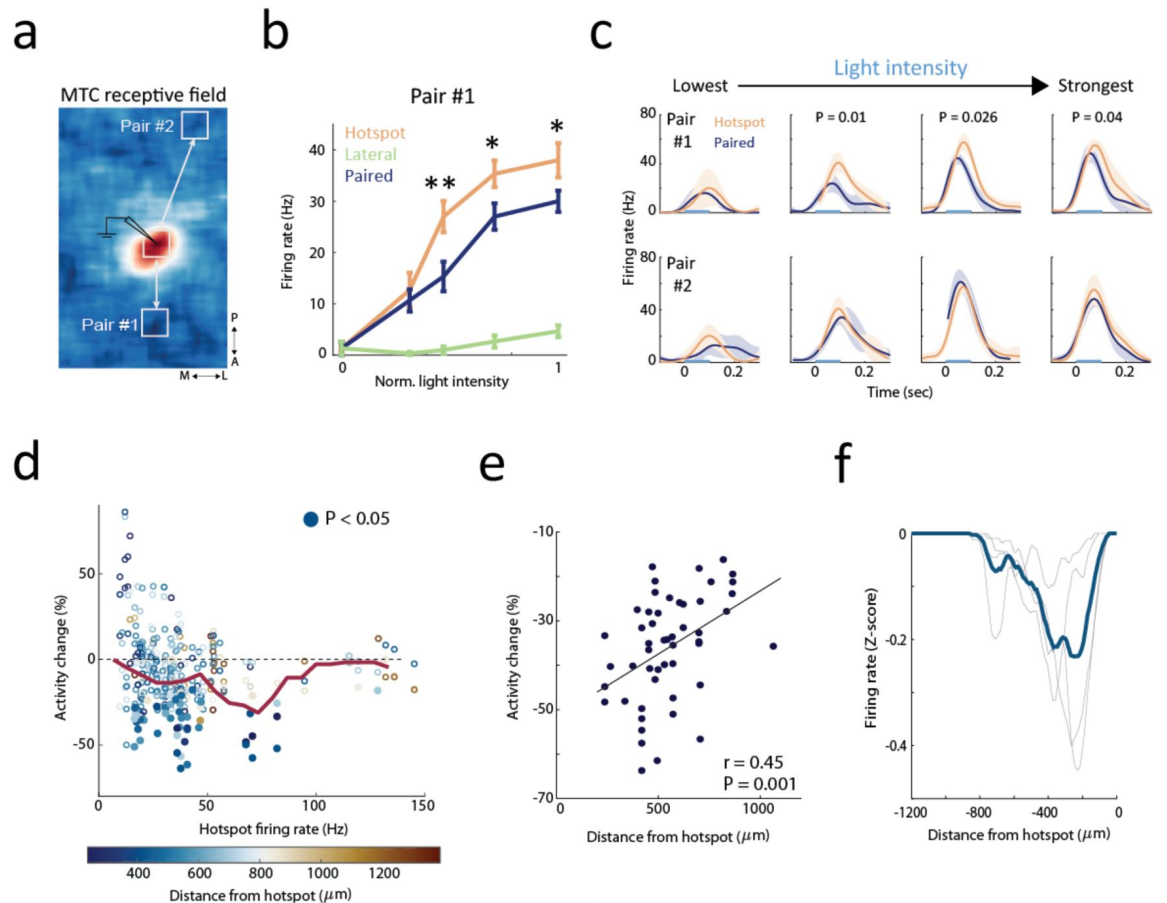
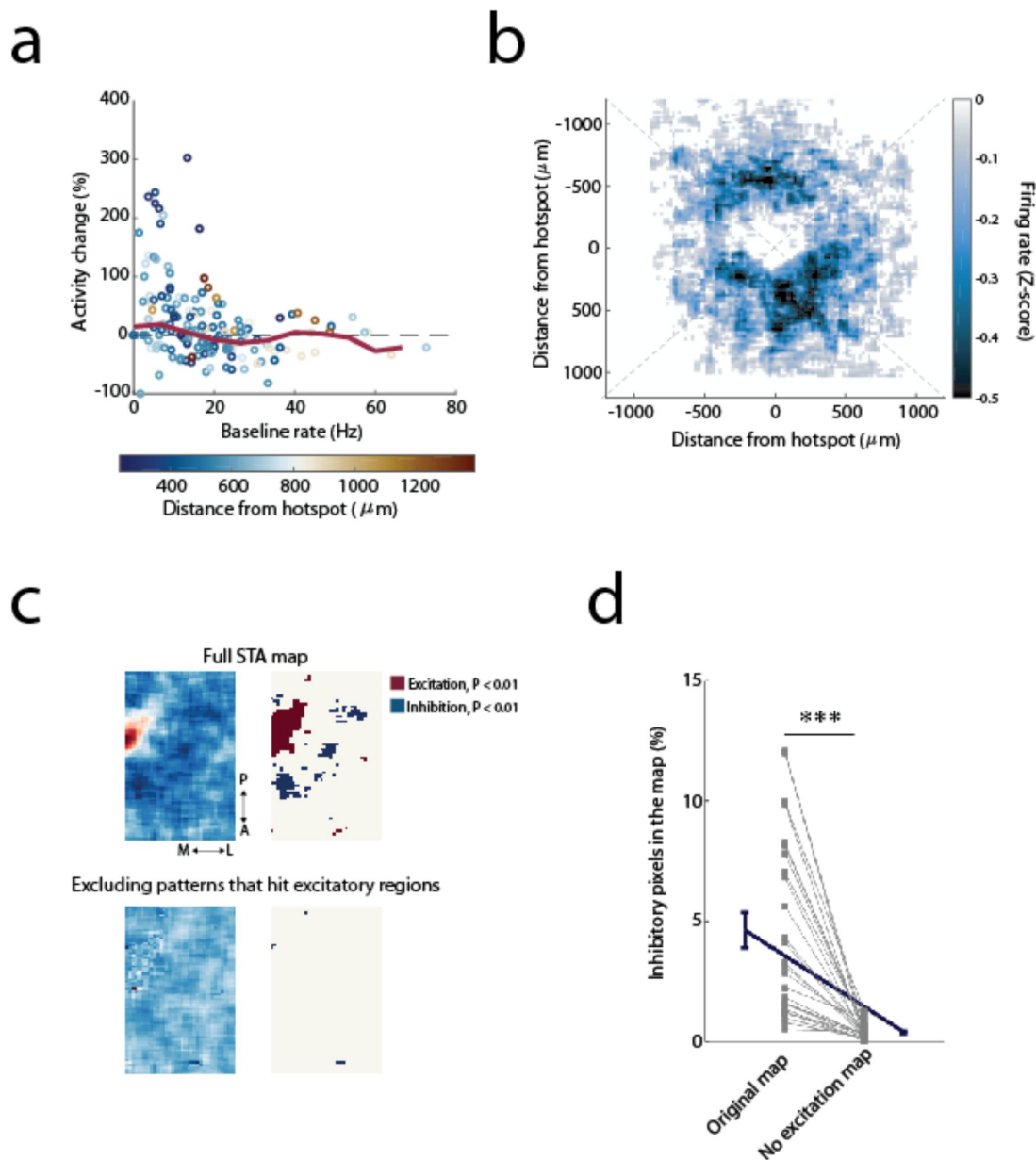


Figure 2

Activity-Dependent Lateral Suppression of MTCs is Confined in Space

- a) An example of an MTC receptive field (STA map). The white rectangles mark the spots exposed to light stimulations. The hotspot location is marked with an electrode drawing.
- b) Mean $\pm$ SEM firing rates of light stimulating pair #1 from **a** for each of the three conditions across all four light intensities. Activation of pair #1 (blue) caused a reduction in the recorded MTC firing rates only when the recorded MTC fired above ~ 25 spikes/sec. Lateral stimulation alone did not inhibit the recorded MTC (green). Zero denotes the baseline firing rate. * $P < 0.05$, ** $P < 0.01$, two-tailed paired t -test.
- c) PSTHs of pair #1 and #2 responses to light stimulations at four different light intensities. Paired stimulation of pair #1 evoked activity-dependent suppression. Paired stimulation of more distant neurons in pair #2 did not affect the recorded MTC firing rates across any tested light intensities. Light stimulation is marked with a blue bar (0.1 sec). PSTHs without a p -value showed no significant change between the firing rates of paired stimulation and hotspot stimulation alone.
- d) Summary analysis of the effect of paired activation on MTCs firing rate. Each point marks the percentage change in firing rate (Y-axis) relative to the firing rate elicited by light stimulating the hotspot alone (X-axis). Suppressive effects (i.e., negative activity change) occurred mainly when the MTC fired in the gamma range (~ 30 – 80 Hz). Color-code denotes the distance between the light-activated MTC pair. Filled circles mark significant activity change ($P < 0.05$, two-tailed unpaired t -test, $N = 51/319$ data points). The red line shows the moving average. Only light intensities that elicited a significant light response were analyzed ($319/511$; $P < 0.05$, two-tailed paired t -test).
- e) Lateral suppression degrades with distance. Spearman correlation between the change in firing rate for all significant inhibitory pairs (the filled circles in **d**, $N = 51$) and their distance to the hotspot ($r = 0.45$, $P = 0.001$).
- f) Mean firing rate of the pixels located on each of the four diagonals in the Z-scored STA maps, centered relative to the 'hotspot'. The centered map and the diagonals are shown in **Supplementary Figure 2b**. Gray lines show the response of the four sections taken from the origin towards the four corners (the four dashed lines in **Supplementary Figure 2b**). The blue line represents the average across all four sections. Zero denotes the hotspot location.



Supplementary Figure 2

MTC lateral suppression is activity- and spatially-dependent

a) Light stimulating a lateral spot without stimulating the hotspot has no effect on the recorded neuron's baseline firing rate, regardless of its firing rates. Color code as in [Figure 2d](#).

b) Lateral inhibition is confined in space. All STA maps were centered at the hotspot location ($N = 27$ Z-scored maps, see Methods).

c) MTC lateral suppression is effective only when the target MTC is activated. Upper panels: an example of an MTC STA map and the corresponding significance map. Lower panels: the same maps recomputed by excluding all light patterns that stimulated the region around the hotspot (All pixels with significant excitation $P < 0.01$, relative to shuffled data). No inhibitory regions are detected after exclusion.

d) Population analysis across all MTC STA maps ($N = 27$) of the percent of inhibitory pixels in the original STA map and when we excluded the light patterns that hit the hotspot area. A pixel is defined as inhibitory if its value is below two standard deviations from the shuffled distribution (see Methods). The percent of inhibitory pixels drops considerably ($P = 1.8\text{e-}6$, two-tailed paired t -test), in the excitation-excluded map.

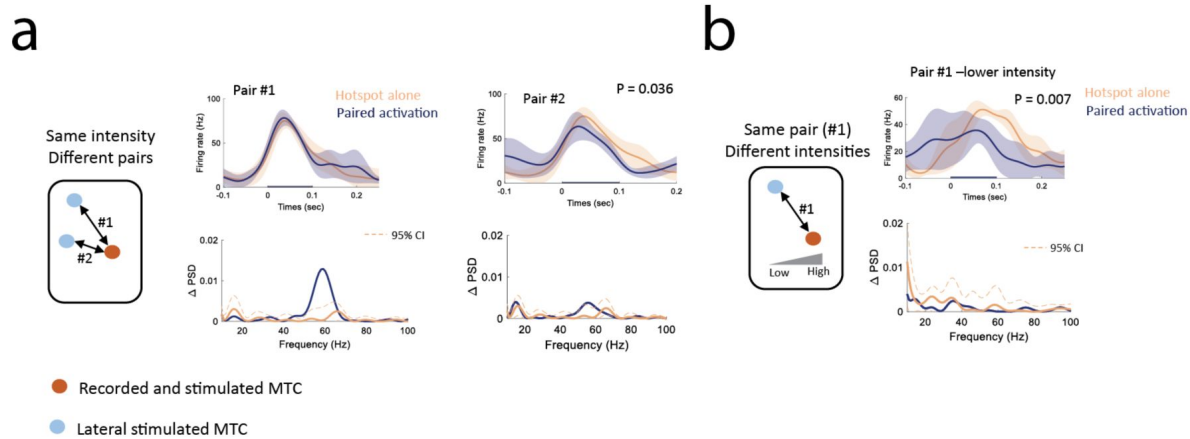


Figure 3

Spike entrainment and suppression are mediated by two different circuits

a) Light-stimulation of two different MTC pairings sharing the same postsynaptic MTC. Light-activating pair #1 (left) caused potent entrainment ($P < 0.05$, two-sample bootstrap) without affecting the light-evoked firing rate ($P = 0.59$, two-tailed paired t -test), whereas light-activating pair #2 (right) suppressed the MTC firing rate ($P = 0.036$, two-tailed paired t -test), without affecting the spikes precision ($P > 0.05$, two-sample bootstrap).

b) Two different light intensities were applied to pair #1, which had differential effects on suppression and entrainment. High light intensity increased spike entrainment without affecting the firing rate (left panel in **a**). In contrast, lower intensity reduced the light-evoked firing rate ($P = 0.007$, two-tailed paired t -test), with no effect on the spikes' gamma entrainment ($P > 0.05$, two-sample bootstrap).

spike in each trial as in (Fukunaga et al., 2014) further showed that GC activation increased the recorded MTC spikes' gamma entrainment (**Figure 4f-g**). Our results strongly suggest that MTC-to-MTC lateral entrainment is mediated by spatially-distributed GCs.

MTC suppression is not mediated by Granule cells

Finally, we examined how GC activation affects MTC odor-evoked firing rate using the same data from the previous experiment (**Figure 4**). We found that light-activating sets of GCs did not change the mean MTC odor-evoked firing rates, irrespective of the MTC firing rate levels. (**Figure 5a**; $N = 54$ light spots tested from 18 cell-odor pairs, $P = 0.64$, two-tailed paired t -test). Moreover, the lack of activity change did not depend on the distance of the activated GCs from the recorded MTC (**Figure 5b**; proximal GCs: $N = 18$, slope = 1, $P = 0.69$; distal GCs: $N = 36$, slope = 0.99, $P = 0.69$, two-tailed paired t -test). These findings suggest that GCs are unlikely to underlie the activity-dependent suppression we observed between MTCs (**Figure 2**). To conclude, our results strongly suggest that MTC-to-MTC lateral entrainment is mediated by spatially distributed GCs, while lateral-suppression, occurring most strongly between adjacent MTCs, is not mediated by GCs.

Discussion

Our findings demonstrate two types of interactions between MTCs that shape their firing rate and temporal dynamics. These two interactions are spatially-dissociated: Lateral entrainment spans the whole accessible bulb surface, while lateral suppression tends to occur primarily between MTCs, activated by adjacent glomeruli. Interestingly, both interactions are activity-dependent, with lateral entrainment and lateral suppression peaking when the postsynaptic MTC fires at ~40Hz and ~70-80Hz, respectively. Furthermore, we found that, in contrast to previous reports, GCs do not mediate MTC-MTC lateral suppression.

MTC and GC interactions facilitate the synchronizing of co-active and spatially-dispersed MTCs

It has been shown that two active MTCs can synchronize their odor-evoked spike timing (Kashiwadani et al., 1999; Schoppa, 2006). However, how this is achieved and how only the odor-activated MTCs are synchronized is unknown. A recent study hypothesized that GCs may support the synchronizing of odor-responding MTCs (Egger and Kuner, 2021). This hypothesis postulates that the spikes of all strongly co-activated MTCs are synchronized to the gamma rhythm, presumably through the GC network. Consistent with this hypothesis, here we have shown that activation of one group of MTCs can increase the synchrony of other, distant, active MTCs, regardless of the distance between them. Most interestingly, we found that this increase in synchronization occurred only if the postsynaptic MTC was firing at a certain level (**Figure 1**), suggesting a simple mechanism by which odor-activated MTCs are synchronized. Furthermore, as predicted, we show that GCs are the interneurons that enable this synchronization.

Lateral suppression is restricted to adjacent and active MTCs

In contrast to the lateral entrainment interactions, we found lateral suppression of MTC spiking is effective mostly between adjacent MTCs and when the postsynaptic neuron is active at a specific range (~30-80Hz, **Figure 2**). However, even when these conditions are met, only ~20% of the tested MTC pairs exhibited significant lateral inhibition. This rate is consistent with previous *in-vitro* studies that found lateral suppression between 10-20% of heterotypic MTC pairs (Isaacson and Strowbridge, 1998; Urban and Sakmann, 2002). These results are also consistent with the increasing evidence that MTC-to-MTC suppression is relatively sparse (Fantana et al., 2008; Lehmann et al., 2016; Pressler and Strowbridge, 2017). The role of this sparse inhibition is

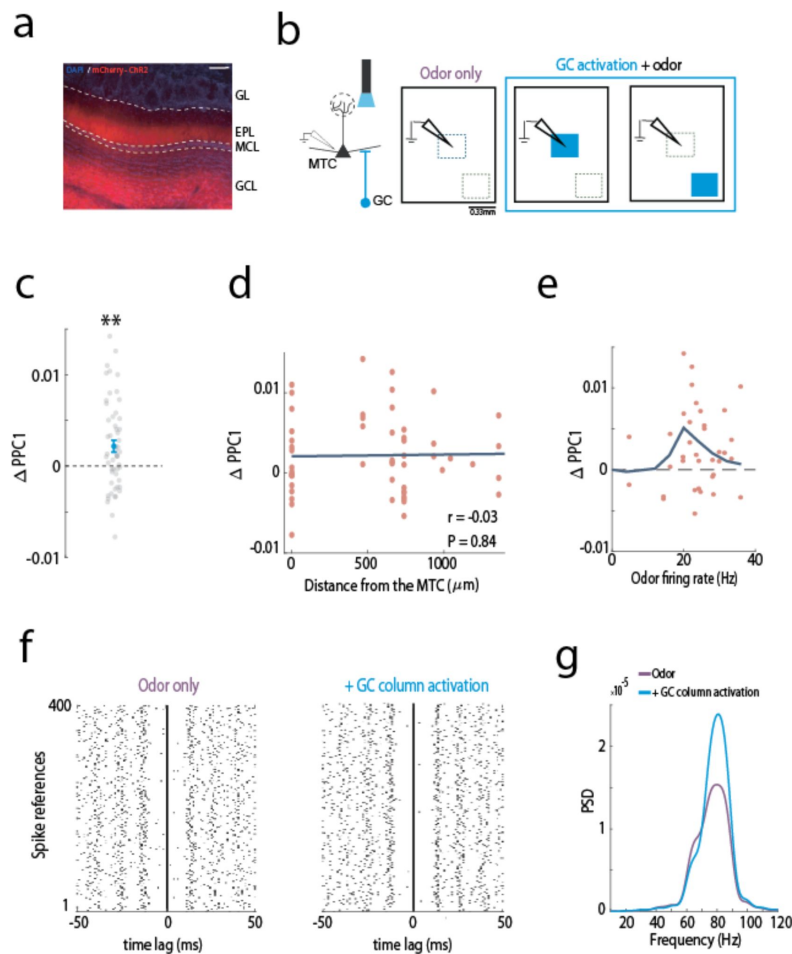


Figure 4

GC activation increases MTC synchrony in an activity-dependent and location-independent manner

a) Cre-dependent AAV injected into the GC layer (GCL) of Gad2-Cre mice. A representative example showing restricted ChR2 expression in the GCL and the external plexiform layer (EPL), into which GCs extend dendrites (red, mCherry-ChR2; blue, DAPI). MCL, mitral cell layer; GL, glomerular layer. Scale bar, 0.1 mm.

b) Schematic illustrations of the experimental setup. Left: Three weeks post injection, MTCs were recorded while light-activating subsets of GCs. Right: MTC activity was recorded in response to odor stimulation alone (purple) or combined with light-activation of GC columns near the recording electrode or distant from it (blue). Scale bar, 330 μ m.

c) MTC spike synchrony to the gamma oscillation (Δ PPC1) significantly increases when we light-stimulated columns of GCs compared to odor-only stimulation ($N = 54$, $P = 0.0016$, two-tailed paired t -test). Only cell-odor pairs that were significantly odor-excited were analyzed ($N = 18/31$ cell-odor pairs; three spots were stimulated per cell odor-pair).

d) MTC spike entrainment does not depend on the GC location. The relation between the change in PPC1 caused by odor and GC stimulation as a function of the distance of the light-stimulated spot from the recording electrode. No significant correlation was found ($N = 54$ values from 18 cell-odor pairs, $r = -0.03$, $P = 0.84$, Spearman correlation). Zero denotes the spot above the recording electrode.

e) MTC spike entrainment is activity-dependent. The change in synchrony peaked when MTCs fired at ~ 25 Hz.

f) Odor-evoked spike reference analysis. Two spike raster plots are shown, for odor only (left, purple), and odor with light-activation of a GC column (right, blue). In each raster plot, spikes are plotted relative to a randomly chosen spike during the odor presentation period ($N = 400$ spikes references, see Methods). Note, the potent spike entrainment when GCs are activated. This analysis was performed on a cell that had a sufficiently high firing rate. This cell is likely a tufted cell due to its potent entrainment at the high gamma range, as shown in (Burton and Urban, 2021; Fukunaga et al., 2014).

g) The power spectral densities (PSD) for the two conditions in f. A multi-taper analysis of the circular convolution of each spike raster plot was used to compute the PSD (see Methods).

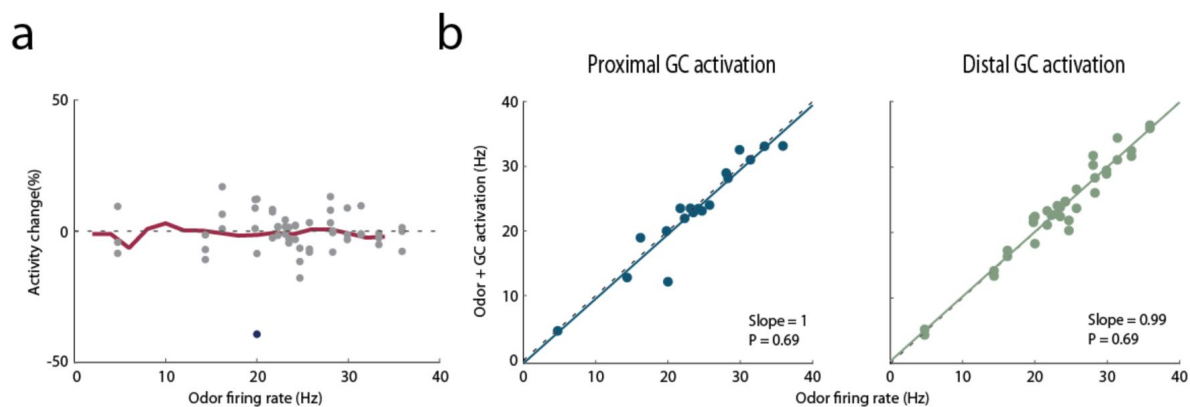


Figure 5

MTC-to-MTC firing rate suppression is not mediated by GCs

a) MTC-to-MTC lateral suppression is not mediated by GCs. Similar to [Figure 2d](#), the change in odor-evoked activity following GC activation is plotted as a function of the recorded MTC odor-evoked firing rates. Filled blue circles denote significant activity change ($P < 0.05$, two-tailed unpaired t -test). A moving average is shown in red.

b) Odor-evoked firing rates are not suppressed when a GC column is activated, irrespective of the odor-evoked MTC firing rate. Two plots are shown for activation of proximal (i.e., in the electrode vicinity, $N = 18$, blue) and distal GCs (all other locations, $N = 36$, green). The slopes of a linear fit and their P -values are plotted in each panel.

unclear. One study suggested that activity-dependent lateral suppression can sharpen MTC odor responses by contrast enhancement (Arevian et al., 2008 [↗](#)). However, a direct experimental evidence is still lacking.

GCs do not substantially affect MTC odor-evoked responses

GCs are the most abundant interneuron type in the OB (Shepherd, 1972), suggesting GCs have a key role in regulating MTC activity. In line with this hypothesis, several key studies have provided evidence that GCs may suppress the MTC firing rate (Arevian et al., 2008 [↗](#); Giridhar et al., 2011 [↗](#); Yokoi et al., 1995 [↗](#)). Here, we show that optogenetically activating arbitrary GC-columns during odor stimulation did not substantially affect MTC odor-evoked firing rates. Although this contradicts several previous in-vitro results, it is consistent with a recent in-vivo study that found no increase in MTC odor-evoked firing rates when silencing GC activity either in anesthetized or in awake mice (Fukunaga et al., 2014 [↗](#)). It is worth noting that light-activating large numbers of GCs can suppress MTC odor-evoked responses (Dalal and Haddad, 2022 [↗](#); Gschwend et al., 2015 [↗](#)). However, activation such as this is unlikely to occur during natural odor response. We speculate that MTC-to-MTC suppression is mediated by EPL interneurons, most likely the Parvalbumin neuron (PV) (Burton, 2017 [↗](#); Kato et al., 2013 [↗](#); Miyamichi et al., 2013 [↗](#)). Future studies are required to shed light on how PV neurons affect MTC activity.

Downstream integration of synchronous MTC activity

How do downstream olfactory cortex neurons benefit from synchronous MTC activity? Anterior piriform cortex (aPC) neurons integrate the activity of several active MTCs and act as coincidence detectors (Davison and Ehlers, 2011 [↗](#); Haddad et al., 2013 [↗](#)). Recently, we have shown that increased MTC spike phase-locking to OB gamma oscillations enhanced aPC neurons' odor representation (Dalal and Haddad, 2022 [↗](#)). This current study extends these findings as it demonstrates that the co-activation of MTCs increases their spikes' gamma entrainment. Moreover, this lateral synchronization is mediated via the GC network. In summary, we found that MTC interactions shape odor-evoked MTC firing rates and spike times. Both of these changes are activity-dependent. Activity-dependent synchronization enables the synchronization of odor-activated MTCs that are dispersed on the OB surface. Such a mechanism is likely to enhance odor information transmission to downstream neurons. On the other hand, activity-dependent spike suppression might enhance odor representation in the OB by reducing low and noisy MTC responses. Finally, our findings suggest that two different OB interneurons mediate these distinct interactions.

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Author Contributions

T.D. performed the experiments and analyzed the data, T.D. and R.H conceptualized the experiments and wrote the paper.

Competing Interest Statement

The authors declare no conflict of interest.

Methods

All surgical and experimental procedures were conducted in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals and Bar Ilan University guidelines for the use and care of laboratory animals in research, and were approved and supervised by the Institutional Animal Care and Use Committee (IACUC). Animals were housed in a group cage and received no experimental treatment, except genotyping. The animals were maintained in a reverse light/dark cycle, and all experiments were performed during the dark period. 19 Tbet-Cre (Jackson Laboratory, Stock No. 024507) crossed with Ai32 (Jackson Laboratory, Stock No. 012569), and 4 Gad2-ires-Cre (Jackson Laboratory, Stock No. 028867) male and female mice aged 3-12 months were used. As no differences in light-evoked responses were observed between sexes, data from both sexes was pooled.

Surgical Procedures for Electrophysiology Recordings

Animals were anesthetized with ketamine/ medetomidine (60/ 0.5 mg/ kg, i.p.) and then fixed in a stereotaxic frame. The bone overlaying the dorsal OBs was removed. Additional anesthesia was administered as needed (~30% of the original dose of ketamine/ medetomidine). Body temperature was maintained at 36-37°C using a homoeothermic blanket system (Harvard Apparatus).

Viral Injections

Mice were briefly anesthetized with isoflurane before intraperitoneal injection of ketamine/ medetomidine (60/ 0.5 mg/ kg i.p.). The mouse was head-fixed in a stereotaxic frame, and AAV5-EF1a-DIO-ChR2-eYFP/mCherry (titers: 4×10^{12} particles/ ml; University of North Carolina Gene Therapy Center) was injected into the granule cell layer. To infect GCs in the GCL, we modified an injection protocol employing five injection sites in two tracks, as reported elsewhere (Fukunaga et al., 2014). The coordinates for the first track were +0.9 mm (M-L) from the midline rhinal fissure, +0.8 mm (A-P), and three sites at the D-V axis at -0.8 mm (300 nl), -1.1 mm (200 nl), and -1.3 mm (200 nl). Second track coordinates were +0.9 mm (M-L) from the midline rhinal fissure, +1.2 mm (A-P) and 2 sites at the D-V axis at -0.8 mm (200 nl), and -1.1 mm (200 nl). M-L coordinates are relative to the midline rhinal fissure. The syringe was left in place for at least three minutes before moving to the next coordinate on the D-V axis. Electrophysiology was carried out at least three weeks post-viral injection. The virus was injected using a micro-injector (IMS-10, Narishige, Japan) at a rate of 70 nl/ min, which was left in place for 5 minutes to allow viral particle diffusion before needle removal. Incisions were closed with tissue glue (Vetbond), and an analgesic injection (Carprofen) was administered at the end of the surgery.

Electrophysiology

The neurons' extracellular activity and local-field potential (LFP) were recorded using tungsten electrodes (~1-10 MΩ; FHC). Neural signals were amplified and first filtered at 1-10,000 Hz and then at 300-5,000 Hz for spiking activity (AM-Systems 1800), sampled, and recorded at 40 kHz (National Instruments, Austin, TX). Spike signals were sorted offline using MClust software in MATLAB (written by A.D. Redish, University of Minnesota). MTC recordings were collected from the dorsal OB (~200-500 μm). The electrode was typically lowered at an angle of 90 degrees. The depth of recorded neurons was estimated when the recording electrode was withdrawn using a micromanipulator.

Optogenetic Stimulation of the Olfactory Bulb

Optogenetic stimulation of neurons in the OB was performed using an optical imaging system based on a digital micro-mirror (OPTOMA X600 DLP Projector), as described (Grobman et al., 2018). Precise spatial control of optical stimulation was achieved by projecting two-dimensional light patterns over the dorsal surface of the OB. We used blue and white light for MTC and GC activation, respectively. For MTC stimulation (Tbet-Cre mice crossed with Ai32 mice), a blue filter was placed on top of the collimating lens, which was placed at a distance of ~20 cm from the projector ($f = 75$ mm, Achromatic doublets, Thorlabs). This configuration resulted in an image where each projected pixel corresponded to a square of $22 \mu\text{m}^2$. The size of the craniotomy determined the light stimulation boundaries of each experiment. Optical stimulation was controlled with the MATLAB psychophysical toolbox. We used a photodiode (FDS1010, 400-ns rise time, Thorlabs) to obtain a timestamp for each light stimulus. The light intensity ranged between ~ 0.1 – 1.5 mW/mm^2 as measured with an optical power meter (Thorlabs PM100D). The stimulation frequency of the optical imaging system was 120 Hz.

Spike-Triggered Average (STA)

The spike-triggered average calculation was described previously by (Grobman et al., 2018). Briefly, STA was used to characterize the receptive field of the recorded MTC. Here, we use the term ‘receptive field’ in a more abstract sense to refer to the ensemble of all neurons on the bulb’s surface that modify the recorded neuron activity.

We stimulated the dorsal OB with patterns of multiple square spots measuring 88 – $110 \mu\text{m}^2$. Light spots within a pattern could overlap up to a shift of one pixel ($22 \mu\text{m}$). The stimuli duration was 0.1 sec, and the inter-stimuli-interval between consecutive patterns ranged between 0.1–0.2 sec. The number of spots projected in each trial ranged between 5–10, and the number of trials per recorded ranged between 2000–3000. We computed the response map for each cell by computing the spike-triggered average, which is the weighted firing rate average of all projected stimuli.

$$STA = \frac{\sum_{i=1}^N P_i R_i}{\sum_{i=1}^N R_i}$$

P_i is the 2-dimensional projected light pattern, and R_i is the evoked firing rate following stimulus P_i .

Paired light stimulation

Single light spots (over the hotspot or a lateral spot) or paired stimulation (both together) of size $\sim 154 \mu\text{m}^2$ were projected at varying light intensities (range ~ 0.1 to $\sim 1.5 \text{ mW/mm}^2$, reported in **Figures 1–3**). The lateral spots were selected based on the STA map or randomly. The number of trials for each condition at a given light intensity was 15, and the inter-stimuli-interval (ISI) was set to 1.5 seconds.

Optogenetic activation of GCs during baseline and odor stimulation

To verify the effect of optogenetic activation of GCs on the recorded MTC, we scanned the OB with light spots, without stimulating the MTC. Light spots were of size $330 \mu\text{m}^2$, each spot was illuminated 20–30 times, the light duration was 0.2 sec, and the ISI was 0.4 sec. The light intensity was as described in (Dalal & Haddad 2022). As expected, we found a ‘coldspot’ (i.e. a reduction in firing rate) in the vicinity of the electrode location (data not shown), confirming that light-activating GCs evokes enough inhibition to reduce MTC baseline activity, consistent with a recent study (Huang et al., 2016). For optogenetic activation of GCs during odor stimulation, the odor duration was 1.5 seconds. The light stimulus was active throughout the entirety of the odor

stimulation period. The activation of proximal and distal GCs during odor stimulation was performed based on the activity map generated under baseline conditions. In each of the three conditions in this experiment, odor stimulation alone or combined with local or lateral GCs activation, the number of trials was at least 10, with an ISI of 10-15 seconds. In these experiments, the LFP was simultaneously recorded with the same electrode, and the signal was later filtered for spiking activity (300-5000 Hz) and LFP ranges (1-300 Hz).

Respiration Analysis

We recorded the respiration signal using a piezoelectric sensor (APS4812B-LW100-R, PUI Audio). The most salient feature of the respiratory signal is the peak in the middle of the inhalation cycle caused by the pressure of the diaphragm on the sensor. The onsets of inhalation and exhalation were defined as the zero-crossings of the signal before and after the peak, respectively.

Odorant Application

Odorants were applied using a custom-built olfactometer. Odorants were diluted in mineral oil (1:100) and stored in sealed glass vials. This concentration was chosen to elicit a detectable response. The tubes were placed in front of the animals' nostrils at a distance of ~2 cm. Airflow was controlled with a mass flow controller (Agilent, Alimc-2LSPM) and set to 0.8 slpm. Air circulated freely between stimulations to reduce odorant remnants. A vent removed residual odorants. Odorant stimulation times and sequences were controlled by custom MATLAB scripts. The odorant stimulation time was set to 1.5 sec., with an inter-trial interval of 10-15 sec. The odorant sequence was randomized, and each stimulus was delivered at least ten times. All odorants used (all at 1% dilution) were from Sigma-Aldrich at their highest purity. The odorants used in the first experiment (reported in [Figure 2](#)) were Phenethylamine (PEA (CAS: 64-41-0), Ethyl acetate (CAS: 141-78-6), ethyl valerate (CAS: 539-82-2), 2-heptanone (CAS: 110-43-0), ethyl butyrate (CAS: 105-54-4), ethyl tiglate (CAS: 5837-78-5), and acetophenone (CAS: 98-86-2).

Quantification and statistical analysis

General Statistical Analysis

All data analyses were performed in MATLAB. The figures or figure legends detail the number of data points used for all statistical tests and graphs. Significance alpha was defined as a 0.05/0.01. Unless stated otherwise, we report the mean $\pm$ SEM or %95 confidence interval. Mean $\pm$ standard deviation is reported when estimated from a bootstrap process. We used a *t*-test as required by the test's null hypothesis and population assumptions. All tests were two-tailed. The test is reported in the figure legend or the main text.

Data Analysis

Spike Sorting

Spike signals were sorted and clustered offline using MClust software in MATLAB (written by A.D. Redish) or Spike3D (Neuralynx). Only visually well-isolated clusters were used, with less than 5% of spikes violating an inter-spike interval of 2 ms.

Significant regions in the STA map

Computation of significant activity change of each pixel of the STA map was done relative to a shuffled STA map. The shuffled map was generated by shuffling the patterns indices and computing the shuffled STA (multiplying the shuffled patterns with the original firing rate as

described above). Then, the 0.5 and 99.5 percentile values from the shuffled map were determined. The values in the STA map that were above or below these low and high percentile thresholds were assigned as significant excitatory and inhibitory pixels, respectively.

Superimposed STA maps

To analyze the spatial location of the inhibitory regions surrounding the recorded MTC, each STA map was Z-scored using the mean and standard deviation values on the map. We then thresholded the map such that only pixels with a Z-score ≤ -1 remained. The hotspot peak was centered at the origin, and the inhibition values were plotted on the map relative to the hotspot. We superimposed all maps and averaged across all neurons.

Exclusion of excitatory regions in STA maps

STA maps were recomputed by excluding trials with at least one light spot that hit the significant excitatory regions in the map. The significance of each pixel in the recomputed map was registered, and the percent of inhibitory pixels in the original and recomputed map was assigned. To verify that the lower number of significant pixels in the recomputed map was not due to a smaller number of trials, for each full map we randomized the same number of trials that were used to construct the recomputed map and computed the STA. We found no significant change in the percent of inhibitory spots compared with the original map (data not shown).

Quantification of MTC lateral inhibition

A pair of light-stimulated MTCs was considered as evoking lateral inhibition if it met two criteria: the response to the hotspot stimulation was significantly different from baseline ($P < 0.05$, one-tailed paired t -test), and there was a significant change in the firing rate across trials between hotspot stimulation and paired activation within a window of 200 ms ($P < 0.05$, two-tailed unpaired t -test, stimulus duration 100 ms). A window of 200 ms was used to account for neural responses with a slower return to baseline. The distance between the spots in the pair was measured using Euclidean distance. The activity change measure was defined as:

$$100 \times \frac{\text{mean}(\text{paired activation firing rate}) - \text{mean}(\text{hotspot firing rate})}{\text{mean}(\text{hotspot firing rate})}$$

MTC entrainment across and within trials

Quantifying MTC spike entrainment across trials was done by computing the PSTH of each condition during the light period (100 ms) using a Gaussian kernel with a standard deviation of 2 ms. The PSTHs were zero-padded to a length of 1 second, Z-scored, and the power spectrum density was computed using multi-taper analysis ($TW = 3$ Hz, $L = (2 \times TW) - 1$, where L is the number of orthogonal Slepian tapers). The delta entrainment value was the difference between the integral of paired activation and the hotspot alone at 40-70 Hz. We analyzed only pairs and light intensities that significantly responded to light stimulation ($P < 0.05$, two-tailed paired t -test). Shuffled distribution of spike entrainment was obtained by shuffling the trial identities and computing the power spectrum density integral difference. Delta entrainment values were labeled significant if they crossed the 95% confidence interval around the shuffled distribution. To quantify the change in spike entrainment within trials, each trial was convolved with a Gaussian window (standard deviation of 2 ms), the power spectrum was computed per trial, and the spectrums were averaged across all trials. The delta entrainment within trials was computed as the sum of the difference between the averaged power spectrums of the two conditions.

Time-Frequency Analyses of MTC entrainment

Time-frequency representation was performed using the continuous wavelet transform (the analytic 'Morse' wavelet, 8 octaves, 32 voices per octave; MathWorks) over the Peri-stimulus time histograms (PSTH) across trials (Gaussian window of 2 ms standard deviation).

Odor-Evoked Responses

An odor firing rate was computed over the first two seconds following odor presentation. This time window included an additional 0.5 sec after stimulus offset to account for neural responses with a slower return to baseline. Trials were aligned to odor onset so that the analyzed neural activity would be aligned with the light stimulation. A response was defined as significant if the firing rate was significantly different from the average firing rate over an equivalent time window prior to stimulus onset ($P < 0.05$, two-tailed paired t -test). PSTHs were smoothed using a Gaussian filter (standard deviation of 50 ms, Chronux toolbox).

LFP Analysis

Recorded LFP signals were down-sampled to 4 kHz and band-pass filtered to the γ -band frequency range (40–70 Hz) using the MATLAB (MathWorks) filter designer (Butterworth IIR filter with order 1). The line noise (50 Hz) was filtered using an IIR bandstop filter (48–52 Hz). We focused our analysis on the low γ -band frequency range as a prominent oscillation in this range in the olfactory bulb was previously reported (Cenier et al., 2009 [↗](#); Lepousez and Lledo, 2013 [↗](#)).

Spikes-LFP pairwise phase consistency (PPC)

Spike-LFP coupling was computed using the pairwise phase consistency 1 (PPC1) measure (Vinck et al., 2012 [↗](#), 2013 [↗](#)). For each neuron, we extracted the spike phases from all trials. To compute the spike phases per trial, we extracted the instantaneous phase of the corresponding γ -band-filtered LFP (40–70 Hz) using the Hilbert transform, assigned a phase to each spike ($0-2\pi$) and pooled the phases across all trials. The PPC1 values were the mean dot-product of all pairwise spike phases, excluding phases of spikes from the same trial.

Spike reference analysis

We performed a spike reference analysis to quantify the entrainment of odor-evoked MTC spikes, as seen elsewhere (Fukunaga et al., 2014 [↗](#)). Spike reference raster plots were constructed for each condition by randomizing 400 spikes from all trials that occurred at a time window of 1 second after odor onset. Each randomized spike served as a reference and was set to time lag 0, and the spikes in a window of ± 50 ms around it were plotted relative to the reference. Choosing a different number of randomized spikes or a different window size did not affect the results. We then computed the PSTH of the raster plot, subtracted its mean, and performed a circular convolution. To compute the power spectral density of the raster plot, we zero-padded the convolved signal into one 1-second length and applied a multi-taper analysis (TW = 5 Hz). The PSD curve was smoothed using a rectangular window of order 15.

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Editors

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Reviewer #1 (Public Review):

Summary:

Dalal and Haddad investigated how neurons in the olfactory bulb are synchronized in oscillatory rhythms at gamma frequency. Temporal coordination of action potentials fired by projection neurons can facilitate information transmission to downstream areas. In a previous paper (Dalal and Haddad 2022, <https://doi.org/10.1016/j.celrep.2022.110693>), the authors showed that gamma frequency synchronization of mitral/tufted cells (MTCs) in the olfactory bulb enhances the response in the piriform cortex. The present study builds on these findings and takes a closer look at how gamma synchronization is restricted to a specific subset of MTCs in the olfactory bulb. They combined odor and optogenetic stimulations in anesthetized mice with extracellular recordings.

The main findings are that lateral synchronization of MTCs at gamma frequency is mediated by granule cells (GCs), independent of the spatial distance, and strongest for MTCs with firing rates close to 40 Hz. The authors conclude that this reveals a simple mechanism by which spatially distributed neurons can form a synchronized ensemble. In contrast to lateral synchronization, they found no evidence for the involvement of GCs in lateral inhibition of nearby MTCs.

Strengths:

Investigating the mechanisms of rhythmic synchronization in vivo is difficult because of experimental limitations for the readout and manipulation of neuronal populations at fast timescales. Using spatially patterned light stimulation of opsin-expressing neurons in combination with extracellular recordings is a nice approach. The paper provides evidence for an activity-dependent synchronization of MTCs in gamma frequency that is mediated by GCs.

Weaknesses:

An important weakness of the study is the lack of direct evidence for the main conclusion - the synchronization of MTCs in gamma frequency. The data shows that paired optogenetic stimulation of MTCs in different parts of the olfactory bulb increases the rhythmicity of individual MTCs (Figure 1) and that combined odor stimulation and GC stimulation increases rhythmicity and gamma phase locking of individual MTCs (Figure 4). However, a direct comparison of the firing of different MTCs is missing. This could be addressed with extracellular recordings at two different locations in the olfactory bulb. The minimum requirement to support this conclusion would be to show that the MTCs lock to the same phase of the gamma cycle. Also, showing the evoked gamma oscillations would help to interpret the data.

Another weakness is that all experiments are performed under anesthesia with ketamine/medetomidine. Ketamine is an antagonist of NMDA receptors and NMDA receptors are critically involved in the interactions of MTCs and GCs at the reciprocal synapses (see for example Lage-Rupprecht et al. 2020, <https://doi.org/10.7554/eLife.63737>; Egger and Kuner 2021, <https://doi.org/10.1007/s00441-020-03402-7>). This should be considered for the interpretation of the presented data.

Furthermore, the direct effect of optogenetic stimulation on GCs activity is not shown. This is particularly important because they use Gad2-cre mice with virus injection in the olfactory bulb and expression might not be restricted to granule cells and might not target all subtypes of granule cells (Wachowiak et al., 2013, <https://doi.org/10.1523/JNEUROSCI.4824-12.2013>). This should be considered for the interpretation of the data, particularly for the absence of an effect of GC stimulation on lateral inhibition.

Several conclusions are only supported by data from example neurons. The paper would benefit from a more detailed description of the analysis and the display of some additional analysis at the population level:

- What were the criteria based on which the spots for light-activation were chosen from the receptive field map?
- The absence of an effect on firing rate for paired stimulations is only shown for one example (Figure 1c). A quantification of the population level would be interesting.
- Only one example neuron is shown to support the conclusion that "two different neural circuits mediate suppression and entrainment" in Figure 3. A population analysis would provide more evidence.
- Only one example neuron is shown to illustrate the effect of GC stimulation on gamma rhythmicity of MTCs in Figures 4 f,g.
- In Figure 5 and the corresponding text, "proximal" and "distal" GC activation are not clearly defined.

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Reviewer #2 (Public Review):

Summary

This study provides a detailed analysis and dissociation between two effects of activation of lateral inhibitory circuits in the olfactory bulb on ongoing single mitral/tufted cell (MTC)

spiking activity, namely enhanced synchronization in the gamma frequency range or lateral inhibition of firing rate.

The authors use a clever combination of single-cell recordings, optogenetics with variable spatial stimulation of MTCs and sensory stimulation *in vivo*, and established mathematical methods to describe changes in autocorrelation/synchronization of a single MTC's spiking activity upon activation of lateral glomerular MTC ensembles. This assay is rounded off by a gain-of-function experiment in which the authors enhance granule cell (GC) excitation to establish a causal relation between GC activation and enhanced synchronization to gamma (they had used this manipulation in their previous paper Dalal & Haddad 2022, but use a smaller illumination spot here for spatially restricted activation).

Strengths

This study is of high interest for olfactory processing - since it shows directly that interactions between only two selected active receptor channels are sufficient to enhance the synchronization of single neurons to gamma in one channel (and thus by inference most likely in both). These interactions are distance-independent over many 100s of μm s and thus can allow for non-topographical inhibitory action across the bulb, in contrast to the center-surround lateral inhibition known from other sensory modalities.

In my view, parallels between vision and olfaction might have been overemphasized so far, since the combinatorial encoding of olfactory stimuli across the glomerular map might require different mechanisms of lateral interaction versus vision. This result is indicative of such a major difference.

Such enhanced local synchronization was observed in a subset of activated channel pairs; in addition, the authors report another type of lateral interaction that does involve the reduction of firing rates, drops off with distance and most likely is caused by a different circuit-mediated by PV+ neurons (PVN; the evidence for which is circumstantial).

Weaknesses/Room for improvement

Thus this study is an impressive proof of concept that however does not yet allow for broad generalization. Therefore the framing of results should be slightly more careful in my opinion.

Along this line, the conclusions regarding two different circuits underlying lateral inhibition vs enhanced synchronization are not quite justified by the data, e.g.

(1) The authors mention that their granule cell stimulation results in a local cold spot (l. 527 ff) - how can they then said to be not involved in the inhibition of firing rate (bullet point in Highlights)? Please elaborate further. In l.406 they also state that GCs can inhibit MTCs under certain conditions. The argument, that this stimulation is not physiological, makes sense, but still does not rule out anything. You might want to cite Aghvami et al 2022 on the very small amplitude of GC-mediated IPSPs, also McIntyre and Cleland 2015.

(2) Even from the shown data, it appears that laterally increased synchronization might co-occur with lateral suppression (See also comment on Figures 1d,e and Figure S1c)

(3) There are no manipulations of PVN activity in this study, thus there is no direct evidence for the substrate of the second circuit.

(4) The manipulation of GC activity was performed in a transgenic line with viral transfection, which might result in a lower permeation of the population compared to the line used for optogenetic stimulation of MTCs.

In some instances, the authors tend to cite older literature - which was not yet aware of the prominent contribution of EPL interneurons including PVN to recurrent and lateral inhibition of MT cells - as if roles that then were ascribed to granule cells for lack of better knowledge can still be unequivocally linked to granule cells now. For example, they should discuss Arevian et al (2006), Galan et al 2006, Giridhar et al., Yokoi et al. 1995, etc in the light of PVN action.

Therefore it is also not quite justified to state that their result regarding the role of GCs specifically for synchronization, not suppression, is "in contrast to the field" (e.g. l.70 f., l.365, l. 400 ff).

Why did the authors choose to use the term "lateral suppression", often interchangeably with lateral inhibition? If this term is intended to specifically reflect reductions of firing rates, it might be useful to clearly define it at first use (and cite earlier literature on it) and then use it consistently throughout.

A discussion of anesthesia effects is missing - e.g. GC activity is known to be reportedly stronger in awake mice (Kato et al). This is not a contentious point at all since the authors themselves show that additional excitation of GCs enhances synchrony, but it should be mentioned.

Some citations should be added, in particular relevant recent preprints - e.g. Peace et al. BioRxiv 2024, Burton et al. BioRxiv 2024 and the direct evidence for a glutamate-dependent release of GABA from GCs (Lage-Rupprecht et al. 2020).

The introduction on the role of gamma oscillations in sensory systems (in particular vision) could be more elaborated.

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Reviewer #3 (Public Review):

Summary:

This study by Dalal and Haddad analyzes two facets of cooperative recruitment of M/TCs as discerned through direct, Chr2-mediated spot stimulations:

- (1) mutual inhibition and
- (2) entrainment of action potential timing within the gamma frequency range.

This investigation is conducted by contrasting the evoked activity elicited by a "central" stimulus spot, which induces an excitatory response alone, with that elicited when paired with stimulations of surrounding areas. Additionally, the effect of Gad2-expressing granule cells is examined.

Based on the observed distance dependence and the impact of GC stimulations, the authors infer that mutual inhibition and gamma entrainment are mediated by distinct mechanisms.

Strengths:

The results presented in this study offer a nice in vivo validation of the significant in vitro findings previously reported by Arevian, Kapoor, and Urban in 2008. Additionally, the distance-dependent analysis provides some mechanistic insights.

Weaknesses:

The results largely reproduce previously reported findings, including those from the authors' own work, such as Dalal and Haddad (2022), where a key highlight was "Modulating GC activities dissociates MTCs odor-evoked gamma synchrony from firing rates." Some interpretations, particularly the claim regarding the distance independence of the entrainment effect, may be considered over-interpretations.

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Author response:

We sincerely appreciate the reviewers' time, effort, and thoughtful feedback, which have significantly contributed to our research.

A key concern raised was the potential overinterpretation of our data. While the reviewers acknowledged our identification of a possible synchronization mechanism among active mitral and tufted cells (MTCs) that is distance-independent, they correctly pointed out that we did not provide direct evidence showing how ensemble MTCs synchronize. We concur with their assessment and will address this in our forthcoming response to ensure a precise interpretation of our findings.

Another concern raised involves the interpretation of results obtained under Ketamine anesthesia. Since Ketamine is an NMDA receptor antagonist, which plays a crucial role in MTC-GC reciprocal synapses, this might impact our conclusions. To address this, we will include analyses demonstrating that optogenetic activation of granule cells (GCs) in an anesthetized state inhibits recorded MTCs during baseline but does not affect odor-evoked MTC firing rates. Additionally, we will thoroughly discuss the potential influence of Ketamine anesthesia on GC-MTC synapses and its implications for our findings.

Lastly, in our detailed response to the reviewers' comments, we will discuss several recent studies that are particularly relevant to our research. We will also expand on our hypothesis that parvalbumin-positive cells in the olfactory bulb may serve as key mediators of the activity- and distance-dependent lateral inhibition observed in our findings.

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